Negation-class walk vs the v2 adding walk: steps per solve on planted toy curves

Study `verification/negation-walk-20261024`. Protocol `PROTOCOL.md` (frozen before any data), `PROTOCOL-AMENDMENT-1.md` and `PROTOCOL-AMENDMENT-2.md` (each written before its arm's data). Frozen run (05:07 to 05:44 UTC, two sittings), amendment 1 (06:42 to 06:49), amendment 2 solves (07:09 to 07:15), rate (08:20 to 08:24), all 2026-10-24, on Intel(R) Xeon(R) Processor @ 2.80GHz (4 vCPU, shared L0 cloud VM, no hardware PMU), g++ (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0. Driver sha256 `f9cacae8d4a37351...`. Every number is read from `results/summary.json` and `results/campaign_saving.json` (measured: steps, ns, operation counters; derived: ratios, intervals, priced costs, projections).

Pre-registered readout (frozen arm B). Verified step reduction at all four primary sizes. Pooled geometric mean of B/A over the four primary sizes: 0.717 (95% CI 0.689 to 0.747); 1/sqrt2 = 0.707. **Amendment 1, arm B2 (post hoc: the escape step that undoes itself, diagnosed from the frozen data, is fixed).** Verified step reduction at all four primary sizes. Pooled geometric mean of B2/A over the four primary sizes: 0.701 (95% CI 0.674 to 0.729); 1/sqrt2 = 0.707. **Amendment 2, arm B3 (post hoc: the escape that rejoins the cycle through a three-step reflection, diagnosed from B2's data, is replaced by c*G; this is the shipped walk).** Verified step reduction at all four primary sizes. Pooled geometric mean of B3/A over the four primary sizes: 0.700 (95% CI 0.673 to 0.728); 1/sqrt2 = 0.707. A/A control passed; positive control (negation walk without cycle handling) passed; reproduction gate (the shipped source reproduces the earlier arms) passed. This is a CPU measurement of the shared walk code on toy curves; the CUDA kernel path was not compiled or run, and no GPU result is claimed.

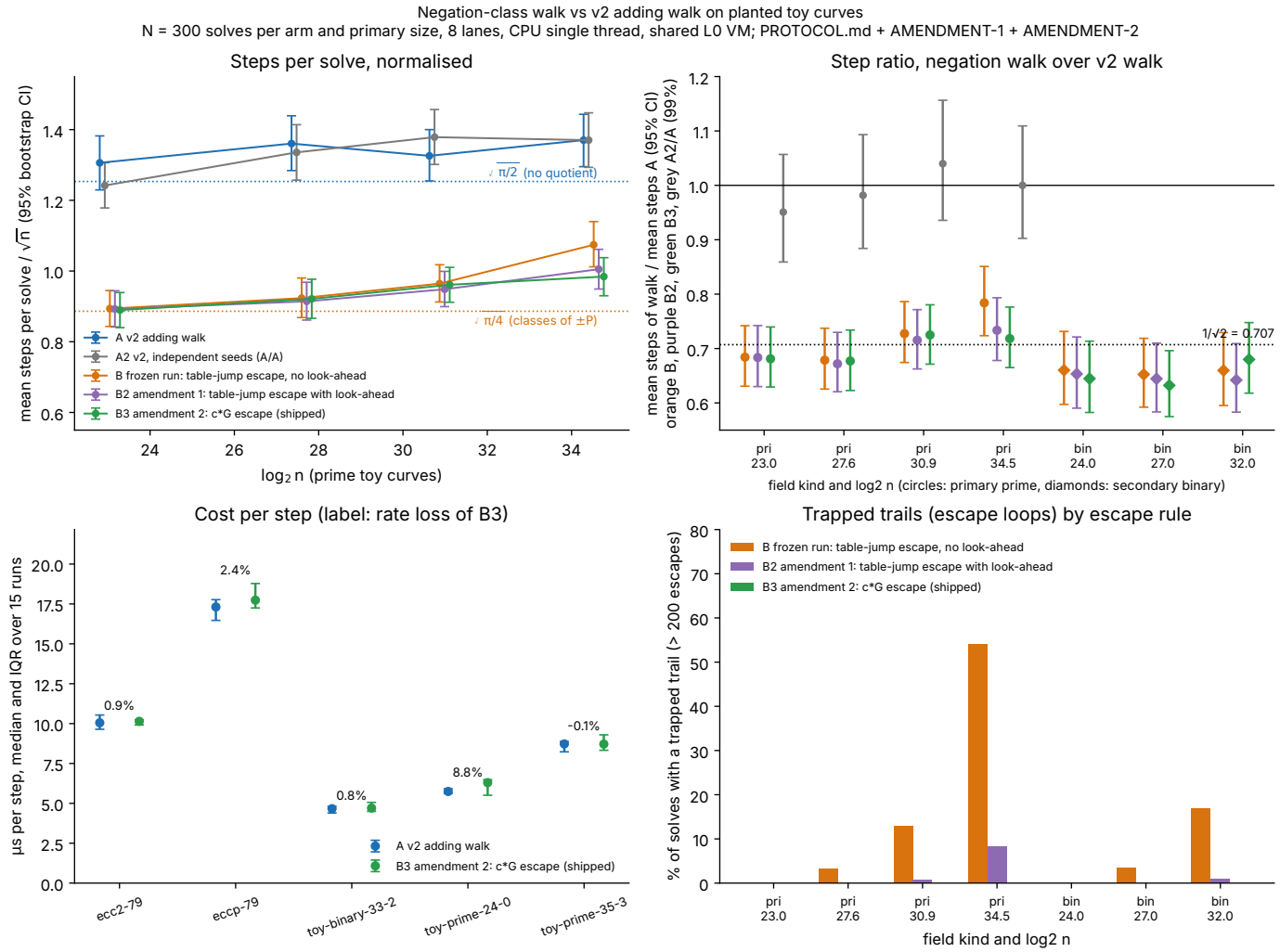


Figure 1: Left: mean steps per solve over sqrt(n), prime toy curves, with the two theory levels. Middle: mean steps of each negation walk over the v2 walk (A) for every measured size (95% bootstrap CI), A/A control in grey (99% CI). Lower left: microseconds per step on each curve (median, IQR of 15 alternating runs) and the rate loss of B3. Lower right: share of solves containing a trapped trail. Source: results/summary.json, drawn by figure.py.

Results (tables.md)

Mean steps per solve (lockstep steps; N solves per arm)

B = walk as first measured (frozen run), B2 = amendment 1, B3 = amendment 2 (the shipped walk).

instance	log2 n	N (A/B/B2/B3)	mean A	mean B	mean B2	mean B3	A / sqrt(pi n/2)	B / sqrt(pi n/4)	B2 / sqrt(pi n/4)	B3 / sqrt(pi n/4)
toy-prime-24-0	23.05	300/300/300/300	3843	2629	2627	2618	1.042	1.008	1.007	1.004
toy-prime-28-1	27.60	300/300/300/300	19374	13149	13019	13119	1.086	1.042	1.032	1.039
toy-prime-31-2	30.87	300/300/300/300	58693	42694	41986	42549	1.058	1.088	1.070	1.085
toy-prime-35-3	34.52	300/300/300/300	215387	168855	157996	154722	1.093	1.212	1.134	1.111
toy-binary-25-0	24.00	200/200/200/200	5389	3558	3521	3474	1.050	0.980	0.970	0.957
toy-binary-28-1	27.00	200/200/200/200	16072	10486	10357	10162	1.107	1.021	1.009	0.990
toy-binary-33-2	32.00	200/200/200/200	91494	60363	58740	62211	1.114	1.039	1.011	1.071

Step ratios to the v2 walk A, 95% bootstrap intervals (1/sqrt2 = 0.707)

instance	B/A (frozen)	B2/A (amend. 1)	B3/A (amend. 2)	covers 1/sqrt2 (B, B2, B3)	B3/B2	A2/A (99%)	C/A
toy-prime-24-0	0.684 [0.631, 0.742]	0.683 [0.630, 0.742]	0.681 [0.629, 0.740]	yes, yes, yes	0.997 [0.919, 1.081]	0.951 [0.859, 1.057]	11.385 [10.020, 12.928]
toy-prime-28-1	0.679 [0.625, 0.737]	0.672 [0.621, 0.730]	0.677 [0.623, 0.734]	yes, yes, yes	1.008 [0.926, 1.094]	0.982 [0.884, 1.093]	76.056 [67.479, 85.448]
toy-prime-31-2	0.727 [0.674, 0.786]	0.715 [0.662, 0.772]	0.725 [0.671, 0.780]	yes, yes, yes	1.013 [0.940, 1.092]	1.040 [0.936, 1.156]	-
toy-prime-35-3	0.784 [0.724, 0.851]	0.734 [0.678, 0.793]	0.718 [0.665, 0.776]	no, yes, yes	0.979 [0.906, 1.058]	1.000 [0.903, 1.109]	-
toy-binary-25-0	0.660 [0.597, 0.731]	0.653 [0.590, 0.721]	0.645 [0.582, 0.714]	yes, yes, yes	0.987 [0.894, 1.088]	-	-
toy-binary-28-1	0.652 [0.592, 0.719]	0.644 [0.583, 0.710]	0.632 [0.575, 0.696]	yes, yes, no	0.981 [0.887, 1.086]	-	-
toy-binary-33-2	0.660 [0.595, 0.729]	0.642 [0.583, 0.709]	0.680 [0.618, 0.748]	yes, yes, yes	1.059 [0.957, 1.174]	-	-

Pooled geometric mean of B/A (frozen) over the four primary sizes: 0.717 [0.689, 0.747]; verified step reduction at all four primary sizes.

Pooled geometric mean of B2/A (amendment 1, post hoc) over the four primary sizes: 0.701 [0.674, 0.729]; verified step reduction at all four primary sizes.

Pooled geometric mean of B3/A (amendment 2, post hoc) over the four primary sizes: 0.700 [0.673, 0.728]; verified step reduction at all four primary sizes.

Controls: A/A pass = True; positive control (C/A lower bound > 2) pass = True; A calibration in [0.8, 1.5] = True; reproduction gate (amended source reproduces the earlier arms) pass = True.

Walk operations and trapped trails

instance	adds/step B / B2 / B3	escapes per 1e6 steps B / B2 / B3	solves with a trapped trail (>200 escapes) B / B2 / B3	trails completed (DPs + cap restarts), B3	trail restarts at the cap per solve A / B / B2 / B3 / C
toy-prime-24-0	1.0158 / 1.0158 / 1.0225	411 / 261 / 265	0 / 0 / 0	24582	0.01 / 0.00 / 0.00 / 0.00 / 38.84
toy-prime-28-1	1.0161 / 1.0161 / 1.0216	1390 / 232 / 229	10 / 0 / 0	30770	0.00 / 0.02 / 0.00 / 0.00 / 354.67
toy-prime-31-2	1.0182 / 1.0161 / 1.0222	2715 / 355 / 257	39 / 2 / 0	24533	0.00 / 0.00 / 0.00 / 0.00 / -
toy-prime-35-3	1.0172 / 1.0166 / 1.0227	8396 / 1031 / 272	162 / 25 / 0	22224	0.00 / 0.01 / 0.00 / 0.00 / -
toy-binary-25-0	1.0160 / 1.0162 / 1.0235	1606 / 307 / 308	0 / 0 / 0	22142	0.03 / 0.03 / 0.00 / 0.00 / -
toy-binary-28-1	1.0158 / 1.0160 / 1.0210	1381 / 205 / 209	7 / 0 / 0	15953	0.00 / 0.00 / 0.00 / 0.00 / -
toy-binary-33-2	1.0156 / 1.0160 / 1.0222	3816 / 474 / 254	34 / 2 / 0	24200	0.00 / 0.03 / 0.00 / 0.00 / -

Long-trail trap check (16 lanes x 2^20 steps, no distinguished points)
Descriptive. Only 16 trails per run: B2's trapped trails start early in a trail (ages 256 to 6656 steps in the solve phase, about 1 in 900 trails at 2^34.5: 25 trapped solves in 22,720 trails), so this check cannot see them (expected 0.02 per run); it shows only that no lane entered a trap later in life. The evidence on trapping is the solve phase above.

curve	arm	steps	escapes per 1e6 steps	lane-rounds with more than 12 escapes (trapped)	max escapes in one lane-round
toy-binary-33-2	B2	16777216	200	0	2
toy-binary-33-2	B3	16777216	191	0	2
toy-prime-35-3	B2	16777216	255	0	2
toy-prime-35-3	B3	16777216	181	0	2

Supplementary per-step timing from the solve phase (post hoc, not the frozen M2)
ns per step = walk time / physical steps over all solves of an arm (two driver processes at a time, arms interleaved in time; A2/A shows the noise level).

instance	A ns/step	A2/A	B/A	B2/A	B3/A
toy-prime-24-0	5347	1.001	1.054	1.051	1.062
toy-prime-28-1	6430	0.997	1.030	1.055	1.068
toy-prime-31-2	7323	1.008	1.031	1.047	1.060
toy-prime-35-3	8131	1.003	1.020	1.046	1.061
toy-binary-25-0	3460	-	1.034	1.023	1.079
toy-binary-28-1	4032	-	1.035	1.004	1.034
toy-binary-33-2	4393	-	1.041	1.063	1.073

B3/A time per step over the 7 instances: mean 1.062, range 1.034 to 1.079.

Priced comparison (group additions + lambda x native operations per solve; walk over A)

instance	arm	ratio at lambda=0	lambda=0.1	lambda=1	break-even lambda	order holds across 0..1
toy-prime-24-0	B	0.695	0.820	1.384	0.28	no
toy-prime-24-0	B2	0.694	0.819	1.383	0.29	no
toy-prime-24-0	B3	0.696	0.822	1.385	0.28	no
toy-prime-28-1	B	0.690	0.814	1.374	0.29	no
toy-prime-28-1	B2	0.683	0.806	1.360	0.31	no
toy-prime-28-1	B3	0.692	0.816	1.376	0.29	no
toy-prime-31-2	B	0.741	0.874	1.475	0.21	no
toy-prime-31-2	B2	0.727	0.858	1.448	0.23	no
toy-prime-31-2	B3	0.741	0.874	1.474	0.21	no
toy-prime-35-3	B	0.797	0.941	1.588	0.15	no
toy-prime-35-3	B2	0.746	0.880	1.485	0.21	no
toy-prime-35-3	B3	0.735	0.867	1.461	0.22	no
toy-binary-25-0	B	0.671	0.792	1.336	0.33	no
toy-binary-25-0	B2	0.664	0.784	1.322	0.34	no
toy-binary-25-0	B3	0.660	0.778	1.312	0.35	no
toy-binary-28-1	B	0.663	0.782	1.320	0.34	no

toy-binary-28-1	B2	0.655	0.773	1.304	0.36	no
toy-binary-28-1	B3	0.646	0.762	1.284	0.38	no
toy-binary-33-2	B	0.670	0.791	1.335	0.33	no
toy-binary-33-2	B2	0.652	0.770	1.299	0.37	no
toy-binary-33-2	B3	0.695	0.820	1.383	0.29	no

Rate of the shipped walk B3 vs A (CPU, single thread, dp_bits=40, 15 alternating runs)

curve	ns/step A (IQR)	ns/step B3 (IQR)	B3/A time per step [95% CI]	rate loss per step	native overhead (GAE/step)	steps B3/A	net time B3/A
toy-prime-24-0	5745 (5628-5914)	6297 (5507-6480)	1.096 [0.954, 1.131]	8.8% [−4.8, 11.6]	0.090	0.681	0.747 [0.600, 0.837]
toy-prime-35-3	8724 (8229-8873)	8716 (8326-9295)	0.999 [0.952, 1.100]	−0.1% [−5.0, 9.1]	−0.017	0.718	0.718 [0.633, 0.854]
toy-binary-33-2	4663 (4392-4808)	4702 (4496-5060)	1.008 [0.955, 1.139]	0.8% [−4.7, 12.2]	−0.007	0.680	0.686 [0.590, 0.852]
eccp-79	17312 (16468-17769)	17736 (17248-18784)	1.025 [0.978, 1.109]	2.4% [−2.3, 9.8]	0.008	-	-
ecc2-79	10051 (9642-10538)	10140 (9914-10242)	1.009 [0.956, 1.056]	0.9% [−4.6, 5.3]	−0.007	-	-

Campaign-level consequence (campaign_saving.md)

Measured constants (mean over the four primary prime sizes; they sit about 7% above theory at these toy sizes (distinguished-point lag, finite walk), so the tables below scale the theoretical E by the measured ratios): v2 mean steps = 1.341 sqrt(n) (theory sqrt(pi/2) = 1.253), shipped negation walk B3 0.939 sqrt(n) (theory sqrt(pi/4) = 0.886); earlier arms B 0.964, B2 0.940 sqrt(n). Steps ratio B3/A 0.700 (B/A 0.719, B2/A 0.701); time per step ratio 1.062 (mean of the seven solve-phase paired timings; the noisier dedicated runs on the 79-bit fields give 1.017); net time ratio 0.744.

campaign	log2 E v2 (sqrt(pi n/2))	log2 E negation (v2 E x measured steps ratio)	steps saved	log2 negation in v2-step time units	note
ecc2-79	39.33	38.81	30%	38.90	
ecc2-89	44.33	43.81	30%	43.90	
ecc2-97	48.33	47.81	30%	47.90	
ecc2k-95	47.83	47.31	30%	47.40	Frobenius follow-up: log2 E = 44.03
ecc2-109	54.33	53.81	30%	53.90	
ecc2k-108	54.33	53.81	30%	53.90	Frobenius follow-up: log2 E = 50.44
ecc2-131	65.33	64.81	30%	64.90	
ecc2k-130	64.83	64.31	30%	64.40	Frobenius follow-up: log2 E = 60.81; production v1 sigma walk log2 E = 60.81 (the negation-only walk is worse here)
ecc2-163	81.33	80.81	30%	80.90	
ecc2k-163	81.33	80.81	30%	80.90	Frobenius follow-up: log2 E = 77.15
ecc2-191	95.33	94.81	30%	94.90	
ecc2-238	118.83	118.31	30%	118.40	
ecc2k-238	118.83	118.31	30%	118.40	Frobenius follow-up: log2 E = 114.38
ecc2-353	176.70	176.19	30%	176.28	
ecc2k-358	179.33	178.81	30%	178.90	Frobenius follow-up: log2 E = 174.58
eccp-79	39.64	39.13	30%	39.21	
eccp-89	44.54	44.03	30%	44.11	
eccp-97	48.58	48.07	30%	48.16	
eccp-109	54.73	54.21	30%	54.30	
eccp-131	65.42	64.91	30%	64.99	
eccp-163	81.50	80.99	30%	81.07	
eccp-191	95.81	95.30	30%	95.39	
eccp-239	119.81	119.29	30%	119.38	
eccp-359	179.56	179.05	30%	179.14	

v2 work done (fraction of E_v2)	v2 remaining (units of E_v2)	negation walk from scratch (steps)	(time)
0.00	1.000	0.700	0.744
0.05	0.952	0.700	0.744
0.10	0.907	0.700	0.744
0.20	0.828	0.700	0.744
0.30	0.759	0.700	0.744
0.50	0.646	0.700	0.744
1.00	0.461	0.700	0.744

Switching a campaign to the negation walk lowers the remaining expected steps only while v2 has done less than 0.40 of its E_v2, and the remaining time while less than 0.32.

Scope and limits

- Steps are counted in lockstep time over 8 lanes (PROTOCOL.md section 4); look-ahead retries and escape moves are steps.
- Timings are L0 CPU timings of generic 12-word bit-serial arithmetic on a shared VM. The frozen rate measurement (M2, single-process alternating runs) is noisy here (IQR 5 to 10%, intervals include 0); the supplementary solve-phase timings (10^6 to 5x10^7 steps per arm and instance, two driver processes, arms interleaved, A2/A within 1%) are tighter and give the per-step ratio used for the campaign table. They are post hoc and labelled so.
- Toy curves (group orders 2^24 to 2^36) say nothing about distinguished-point overhead at production DP densities, and nothing about GPU rates.
- A negation-walk namespace starts empty: its points never combine with v2 points, so v2 work already done on a campaign is not credited.